

PULSE COUNTER USING AT89C4051

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Pulse counters are widely used in our day-to-day life. Almost all museums and theatres have a visitor counter installed at entry/exit to measure the visitor traffic. In industries, counting is done for production control. Market tests too are performed by counting the sold goods.

A pulse counter could be roughly divided into three parts: a pulse source, an electronic device that counts, stores and prepares outputs, and a display to show the accumulated count.

This pulse counter is based on Atmel AT89C4051 microcontroller. TTL-logic-compatible pulses generated by the source are fed to the counter for counting. The AT89C4051 is a low-voltage, high-performance, 8-bit microcontroller with 4 kB of Flash programmable and erasable read-only

memory, 128 bytes of RAM, 15 input/output (I/O) lines, two 16-bit timers/counters, a five-vector, two-level interrupt architecture, a full-duplex serial port, a precision analogue comparator, on-chip oscillator and clock circuitry.

System clock plays a significant role in operation of the microcontroller. An 11.0592MHz quartz crystal provides basic clock to the microcontroller (IC1) at its pins 4 and 5. Power-on reset is provided by electrolytic capacitor C3 and resistor R1. Switch S1 is used for manual reset.

Port pin P3.5 receives the input pulse and the count is displayed on the LCD as well as hyper terminal of the PC. Pulse reception is indicated by LED2, which is connected to port pin



PARTS LIST

Semiconductors:

- IC1 - AT89C4051 microcontroller
- IC2 - MAX232 RS-232 driver
- IC3 - 7805, 5V regulator
- D1-D4 - 1N4007 rectifier diode
- LED1, LED2 - 5mm LED
- LCD - 16-character × 2-line

Resistors (all ¼-watt, ±5% carbon):

- R1 - 10-kilo-ohm
- R2, R3 - 330-ohm
- R4 - 33-ohm

Capacitors:

- C1 - 1000µF, 25V electrolytic
- C2 - 0.1µF ceramic disk
- C3 - 10µF, 16V electrolytic
- C4, C5 - 33pF ceramic disk
- C6-C10 - 1µF, 16V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V, 500mA secondary transformer
- S1 - Push-to-on switch
- X_{1AL} - 11.0592MHz crystal
- 9-pin D-type female COM port connector
- Heat-sink

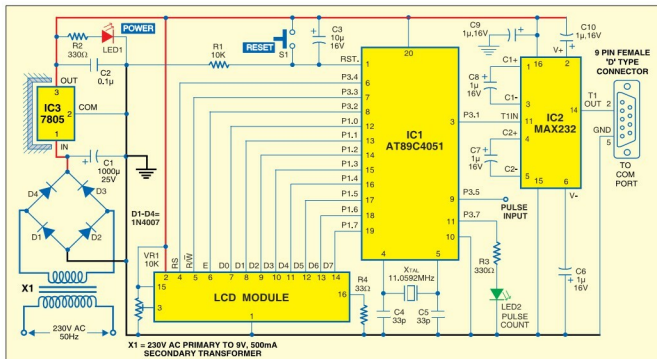


Fig. 1: Circuit of microcontroller-based pulse counter